Visualization Draft

1. Set up

Packages

```
# general use (data wrangling and plotting)
library(tidyverse)

# clean column names
library(janitor)

# file organization
library(here)
```

Data

```
# weather data from NOAA weather station
weather <- read_csv(here("data", "NOAA-weather-data.csv"))

# water quality survey metadata
metadata <- read_csv(here("data", "NCOS_YSI_Water_Quality_Monitoring_0.csv"))

# water quality parameter measurements
water_parameters <- read_csv(here("data", "YSI_Data_Begin_1.csv"))
```

Cleaning weather

```r
# creating new object from weather
weather_clean <- weather |>
  # cleaning column names
  clean_names() |>
  # making sure date is in POSIXct format
  mutate(date = mdy(date)) |>
  # selecting columns of interest
  select(date, prcp, tmax, tmin) |>
  # extracting month and year from date column
  mutate(month = month(date),
         year = year(date)) |>
  # assigning water year
  mutate(wy = case_when(
    # when month is October or later, assign value of year + 1
    month >= 10 ~ year + 1,
    # when month is September or earlier, assign value of existing year
    .default = year))
```

Cleaning metadata

```r
# creating new object from metadata
metadata_clean <- metadata |>
  # cleaning column  names
  clean_names() |>
  # excluding unnecessary columns
  select(!c(object_id, creator, editor)) |>
  # making sure monitoring_date is in POSIXct format
  mutate(monitoring_date = mdy_hms(monitoring_date)) |>
  # renaming monitoring datetime column to reflect values
  rename(monitoring_datetime = monitoring_date) |>
  # creating new columns for month and year of observations
  mutate(month = month(monitoring_datetime),
         year = year(monitoring_datetime)) |>
  # assigning water year
  mutate(wy = case_when(
    # when month is October or later, assign value of year + 1
    month >= 10 ~ year + 1,
    # when month is September or earlier, assign value of existing year
    .default = year)) |>
  # create new column of full site name
```

```r
  mutate(site_full = case_when(
    site_name == "east_channel" ~ "East Channel",
    site_name == "phelps_bridge" ~ "Phelps Bridge",
    site_name == "venoco_bridge" ~ "Venoco Bridge",
    site_name == "other" ~ "Other"
  ))
```

Cleaning water parameters

```r
# creating new object from water_parameters
water_parameters_clean <- water_parameters |>
  # cleaning object names
  clean_names() |>
  # excluding unnecessary columns
  select(!c(creator, editor)) |>
  # renaming columns
  rename(elevation_code = depth_code) |>
  # replacing incorrect elevation code from 2021-07-16
  mutate(elevation_code = case_when(
    elevation_code == 10 ~ 1,
    .default = elevation_code
  )) |>
  # set factors
  mutate(elevation_code = as_factor(elevation_code),
         elevation_code = fct_relevel(elevation_code,
                                      "0", "1", "2", "3", "4", "5", "6"))
```